



Hades

A Generalist Vision-Language-Action Model for Long-Horizon Real-Time GUI Game Agents

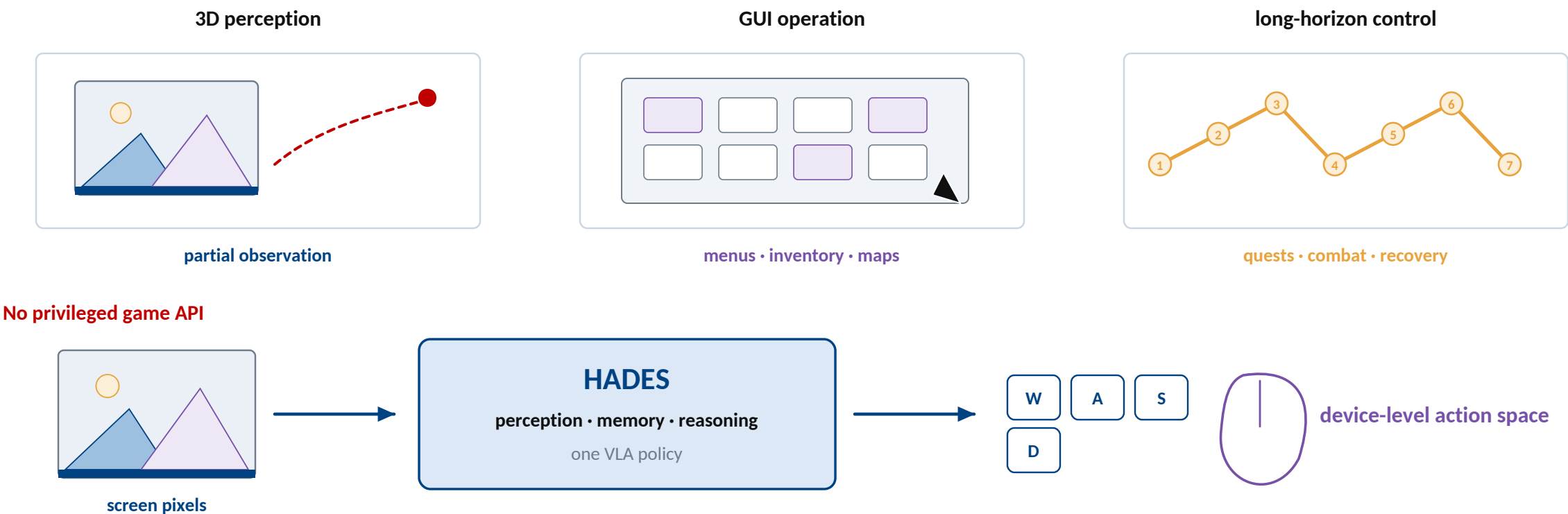


miHoYo Lumi · full author list in the paper's Contributions section

Industrial-scale generalist agent research

Manuscript dated 26 August 2026

Commercial games expose the full agent problem

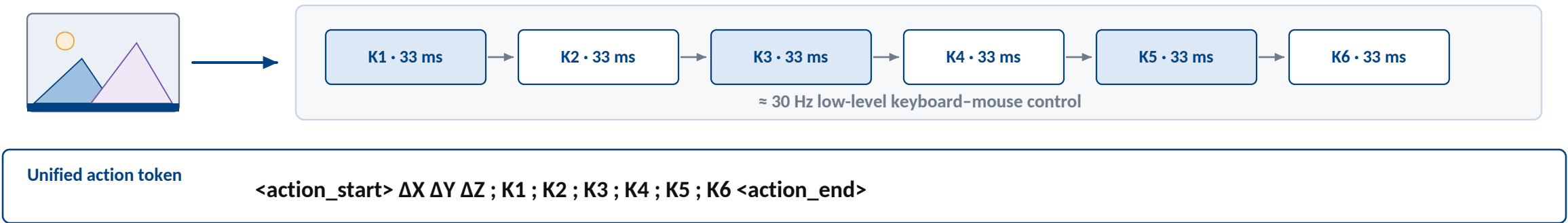


Pixels in, keyboard and mouse out — with **hours of state** in between.

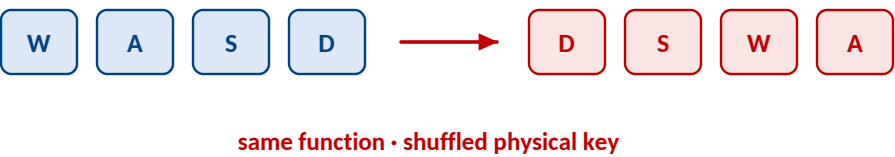
Stage 1 — learn a universal device-level action language

Decouple visual and action time scales

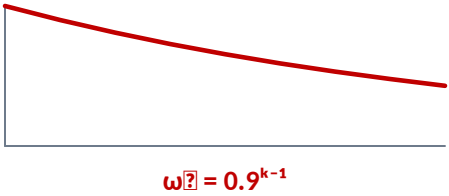
one decision window = 200 ms



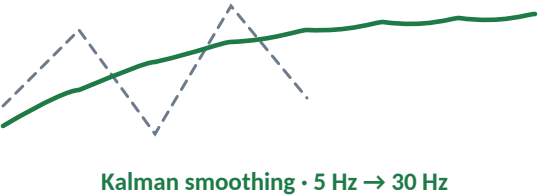
Key Shuffle



Time-aware loss decay



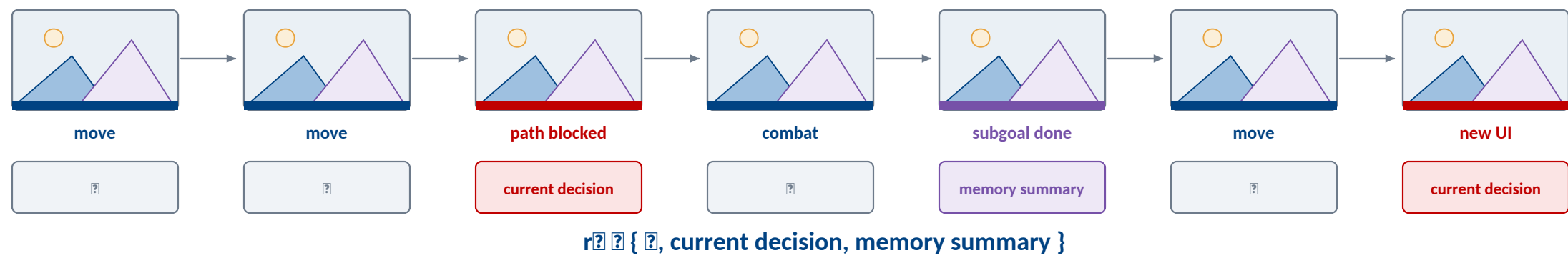
Mouse inversion



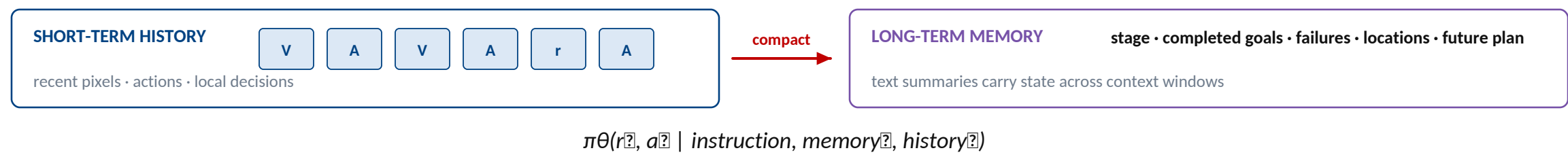
Compress 30 Hz control into tokens, then debias what the model learns from repetition.

Stage 2 — think sparsely and remember across context windows

Most steps execute; a few steps decide



Two timescales of memory



Routine control stays fast; decisions and memory appear only at **critical moments**.

Stage 3 — TA-GRPO turns an hour-long rollout into learnable segments

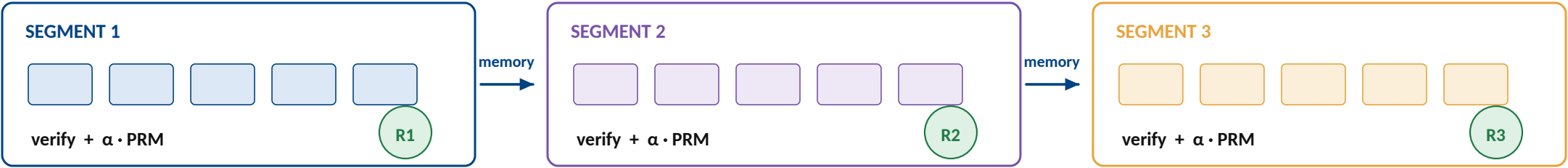
Naïve rollout-level RL

context overflow • sparse terminal reward



one minute already exceeds typical language-task scale

TA-GRPO: segment, remember, reward locally



Time-aware credit inside each segment

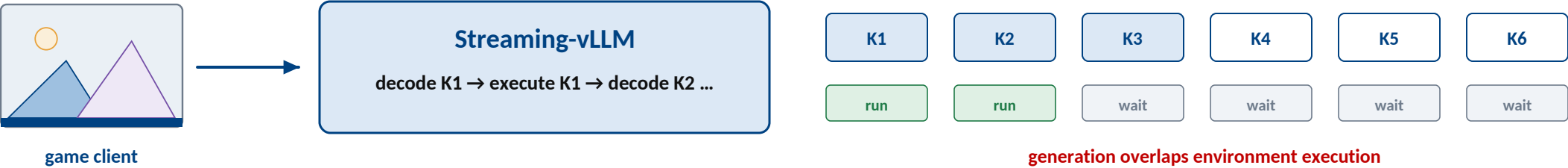


$$A_{g,t} = \beta^{T-g-1-t} A_g$$

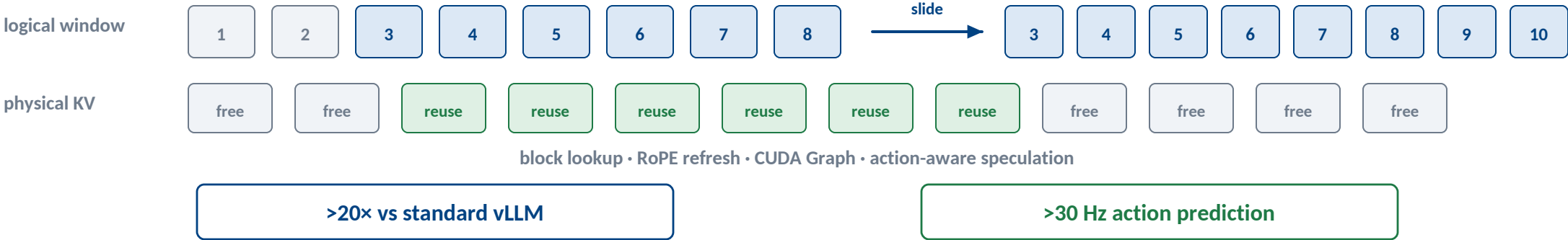
Compact memory across segments and assign credit near the state changes that matter.

Deployment — stream actions while the model is still decoding

Streaming action generation



Long context without recomputation

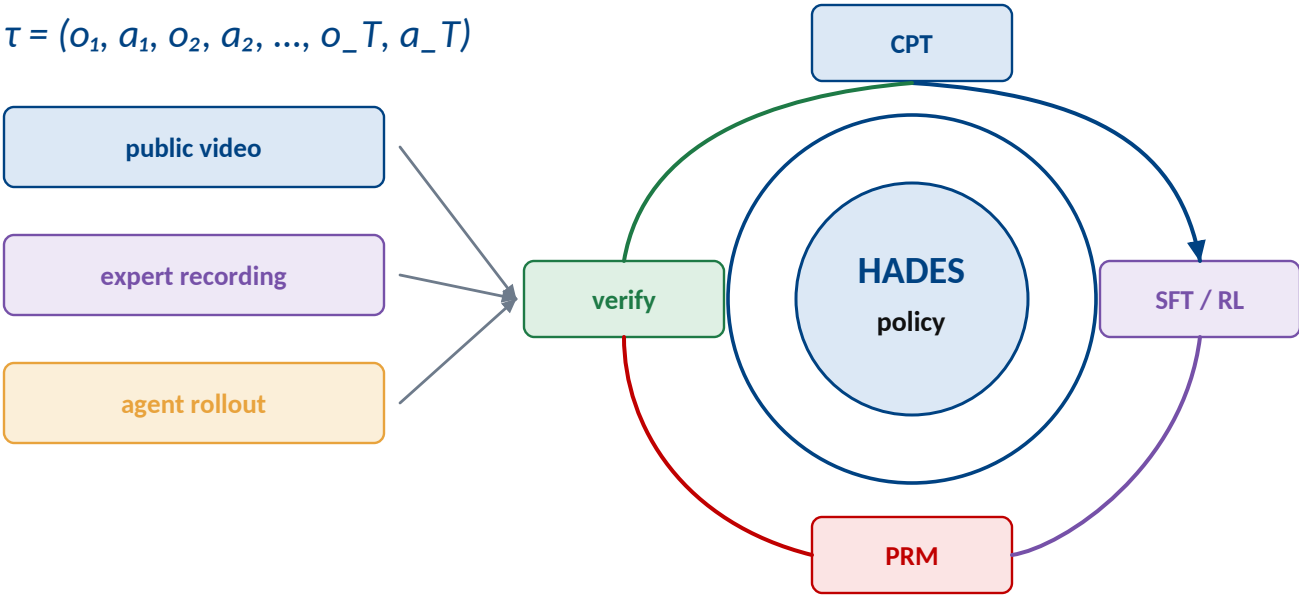


Streaming-vLLM overlaps generation with execution and reuses the moving context.

System view — data, policy, verifier, and infrastructure form one loop

Unified raw format

$$\tau = (o_1, a_1, o_2, a_2, \dots, o_T, a_T)$$



Industrial execution loop



100+ parallel workers

checker + model judger

MATERIAL BOUNDARY

the supplied PDF states near-human / SOTA results but includes no benchmark tables to verify them

The contribution is an industrial agent stack, not only a policy model.